7. SURFACE TENSION BY CAPILLARY RISE METHOD

AIM

To determine surface tension of a liquid by capillary rise method.

APPARATUS REQUIRED

A beaker of Water, capillary tube, vernier microscope, double hole rubber stopper, a knitting needle, a short rubber tubing and retort clamp.

FORMULA

The surface tension of the liquid $T = \frac{hr\sigma g}{2}$ $N\,m^{-1}$

where $T \leftarrow$ Surface tension of the liquid $(N\,m^{-1})$

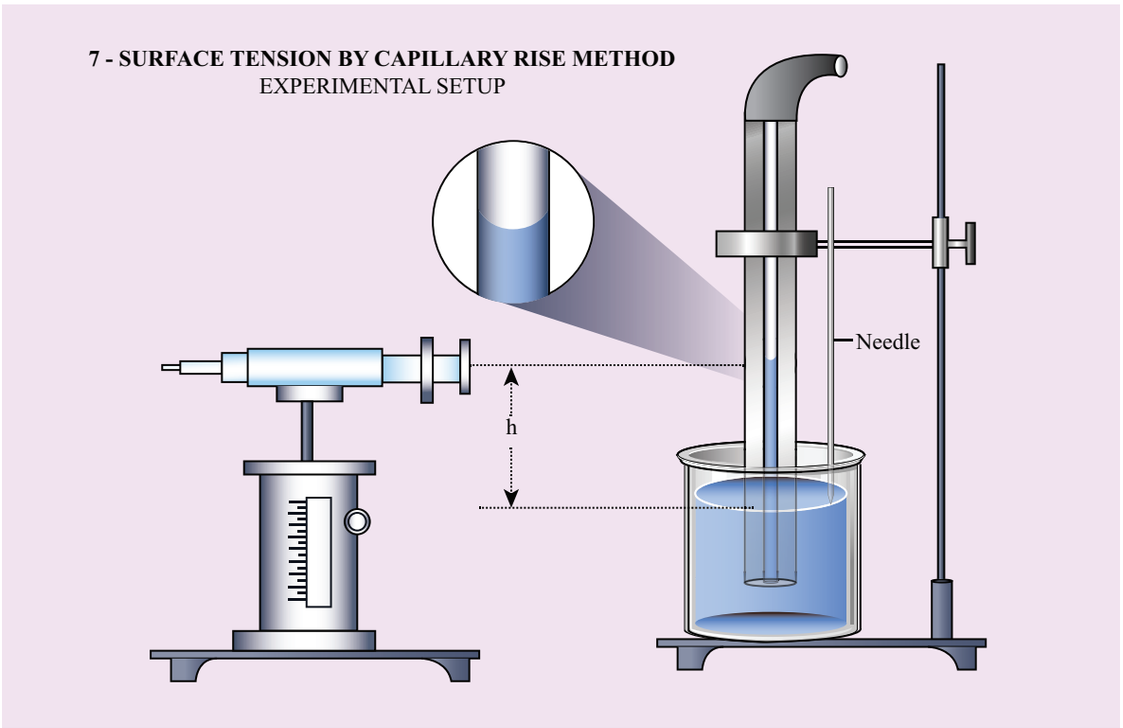
$h \leftarrow$ height of the liquid in the capillary tube (m)

$r \leftarrow$ radius of the capillary tube (m)

$\sigma \leftarrow$ Density of water $(kg\,m^{-3})$ $(\sigma = 1000\ kg\,m^{-3})$

$g \leftarrow$ Acceleration due to gravity (g = 9.8 m s^{-2})

DIAGRAM



PROCEDURE

- A clean and dry capillary tube is taken and fixed in a stand
- A beaker containing water is placed on an adjustable platform and the capillary tube is dipped inside the beaker so that a little amount of water is raised inside.

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 297 02-01-2023 15:28:03